PixelCNN & Its Improvements

Understanding Autoregressive Image Generation with Masked Convolutions

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Outline

1. Background: Autoregressive Image Generation
2. PixelCNN: Masked Convolutions
3. Receptive Field & Sawtooth Pattern
4. The Blind Spot Problem
5. Solution: Gated PixelCNN
6. Summary

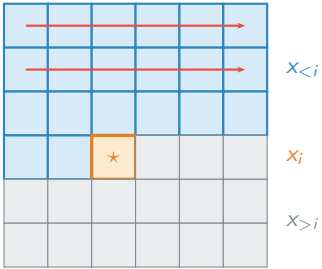
Autoregressive Factorization

Model the joint distribution of all pixels by the **chain rule**:

$$p(\mathbf{x}) = \prod_{i=1}^{n^2} p(x_i \mid x_1, \ldots, x_{i-1})$$

Key design choices:

- **Ordering**: raster scan (left-to-right, top-to-bottom)
- **Conditional model**: a neural network
- **Challenge**: how to enforce that $p(x_i|\cdot)$ only depends on *previous* pixels?



Raster scan order

PixelCNN: Core Idea

Insight: Use masked convolution kernels so that each output pixel only depends on previous pixels in the raster scan order.

Standard convolution:

$$y_{r,c} = \sum_{i,j} w_{i,j} \cdot x_{r+i,c+j}$$

Masked convolution:

$$\boxed{y_{r,c} = \sum_{i,j} \underbrace{m_{i,j}}_{\text{mask}} \cdot w_{i,j} \cdot x_{r+i,c+j}}$$

The mask m zeroes out weights that would access

Standard Conv Kernel

w	w	w	w	w
w	w	w	w	w
w	w	w	w	w
w	w	w	w	w
w	w	w	w	w

All positions active

$\downarrow$ Apply mask m $\downarrow$

Masked Conv Kernel

w	w	w	w	w
w	w	w	w	w
w	w	0	0	0
0	0	0	0	0

The 5×5 Mask in Detail

Mask m (from the assignment)

Full rows	1	1	1	1	1
	1	1	1	1	1
Center	1	1	0	0	0
Zeroed	0	0	0	0	0
	0	0	0	0	0

Center pixel

Three distinct regions:

1. **Upper rows** (rows 0–1):
   All 5 columns = **1**
   $\Rightarrow$ See *full width* above
2. **Center row** (row 2):
   Only left 2 columns = **1**
   $\Rightarrow$ See only *left neighbors*
3. **Lower rows** (rows 3–4):
   All columns = **0**
   $\Rightarrow$ *Completely invisible*

Key Asymmetry

Upper rows: left **and** right visible.
Center row: **only left** visible.

Type A vs Type B Masks

Type A Mask (1st layer)

1	1	1	1	1
1	1	1	1	1
1	1	0	0	0
0	0	0	0	0
0	0	0	0	0

Center pixel is **excluded** (0).
Used in the **first** layer only —
the pixel cannot see *itself*.

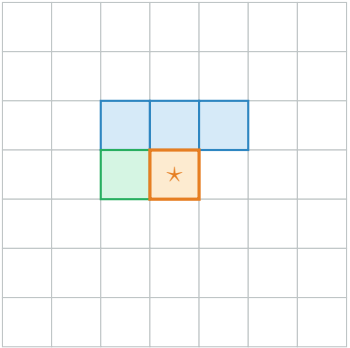
Type B Mask (layers 2+)

1	1	1	1	1
1	1	1	1	1
1	1	1	0	0
0	0	0	0	0
0	0	0	0	0

Center pixel is **included** (1).
Used in **subsequent** layers —
features at the same position carry
information from prior pixels only.

Both masks preserve the autoregressive property: x_i never depends on x_j for $j \geq i$ in the input.

Receptive Field After 1 Layer (3×3 kernel)



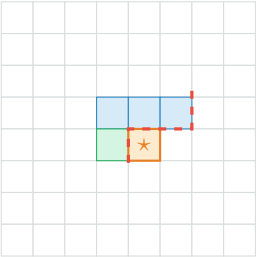
With a 3×3 masked kernel ($h = 1$):

- **Row above**: sees columns $c-1, c, c+1$ (3 pixels, full width)
- **Same row**: sees only $c-1$ (1 pixel, left only)
- **Below**: nothing

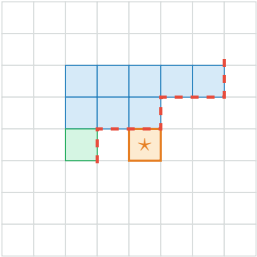
RF size: **4 pixels**.
Right boundary: $c+1$ above, $c-1$ on same row.

Receptive Field Growth: 1, 2, 3 Layers

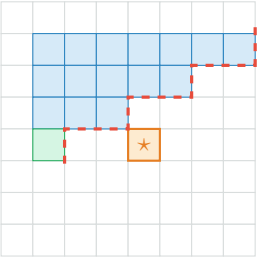
1 Layer



2 Layers

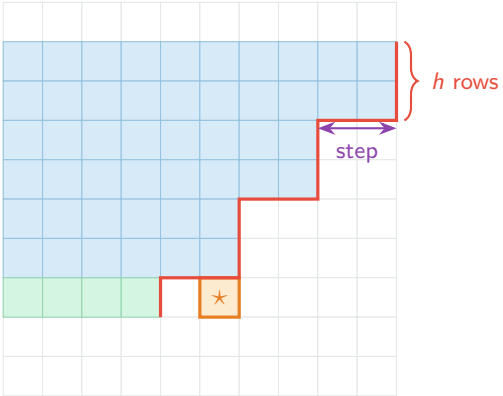


3 Layers



The **red dashed line** is the right boundary — it forms a staircase / sawtooth shape. Each row going down, the right boundary shifts left. Upper rows expand both ways; current row only expands left.

The Sawtooth Pattern Explained



For a 5×5 kernel ($h = 2$):

- The right boundary stays **flat** for $h = 2$ rows (a "tooth")
- Then it **drops** by h columns
- Repeat: flat $\rightarrow$ drop $\rightarrow$ flat $\rightarrow$ drop

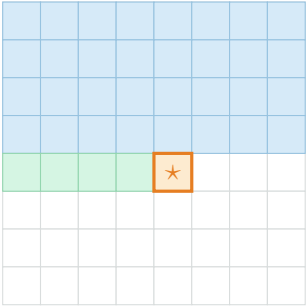
Why?

- The mask has $h = 2$ full rows above center $\Rightarrow$ each layer adds h rows of symmetric expansion
- On the center row, only leftward expansion
- The periodic "h flat + step" pattern creates the sawtooth

Consequence

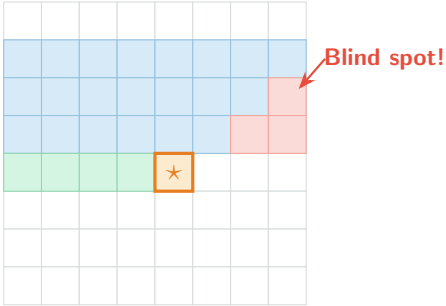
Ideal RF vs Actual RF: The Blind Spot

Ideal Receptive Field
(all previous pixels in raster order)



Everything above & left

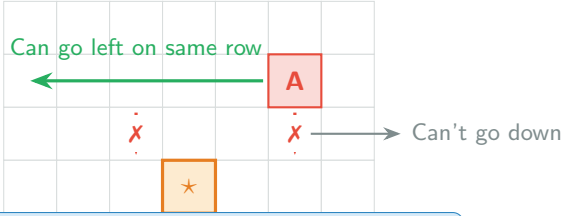
Actual Receptive Field
(after stacking masked conv layers)



Upper-right region is missing

The blind spot = upper-right pixels that should be visible but are not reachable by stacked masked convolutions.

Why Does the Blind Spot Exist?

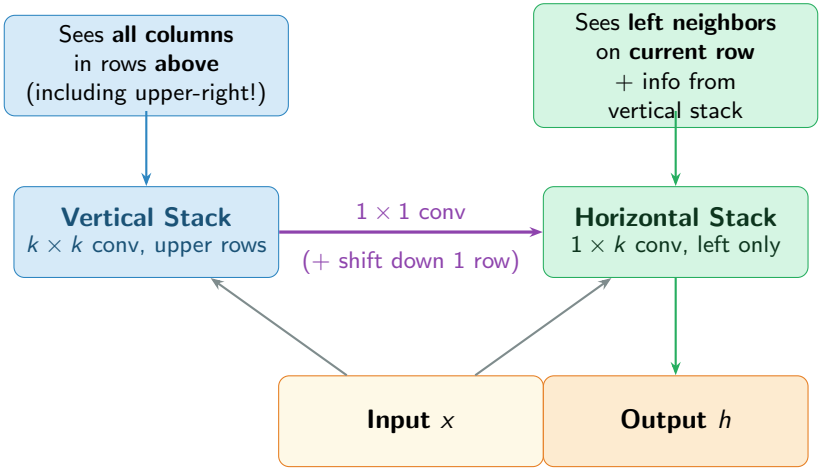


On the **center row**: mask only allows **leftward** flow. **Below** center row: mask is all zeros — **no flow at all**.

$\Rightarrow$ Pixel A's information has **no path** to reach $\star$.

The issue: information on row $r-k$ at column $c+j$ must first travel *down* to row r, but the mask blocks all downward propagation except through the upper rows of the next layer's kernel — which shifts the right boundary left.

Key Idea: Split into Two Stacks



Core Insight

Separate "looking up" (vertical) from "looking left" (horizontal), then **connect** them. The

Vertical Stack: See Everything Above

Vertical Mask (5×5)

1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
0	0	0	0	0
0	0	0	0	0

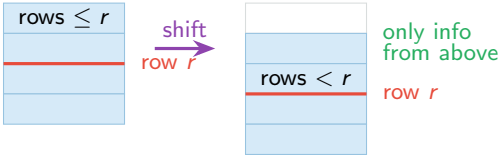
Top 3 rows: **all** columns active.
Center row included, bottom rows zero.

Computation:

$$v = \mathrm{conv}(m_v \cdot w_v,\ x)$$

Critical step: shift output down by 1 row!

Before shift **After shift**



After shift: position (r, c) in v only contains information from rows **strictly above** r.
$\Rightarrow$ Autoregressive property preserved!

Horizontal Stack + Vertical Connection

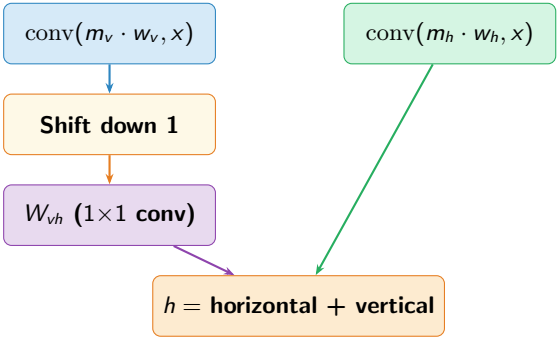
Horizontal Mask (1×5)

1	1	1	0	0

Center (included in Type B)

Only sees **current row**, pixels to the **left** (and self for Type B).

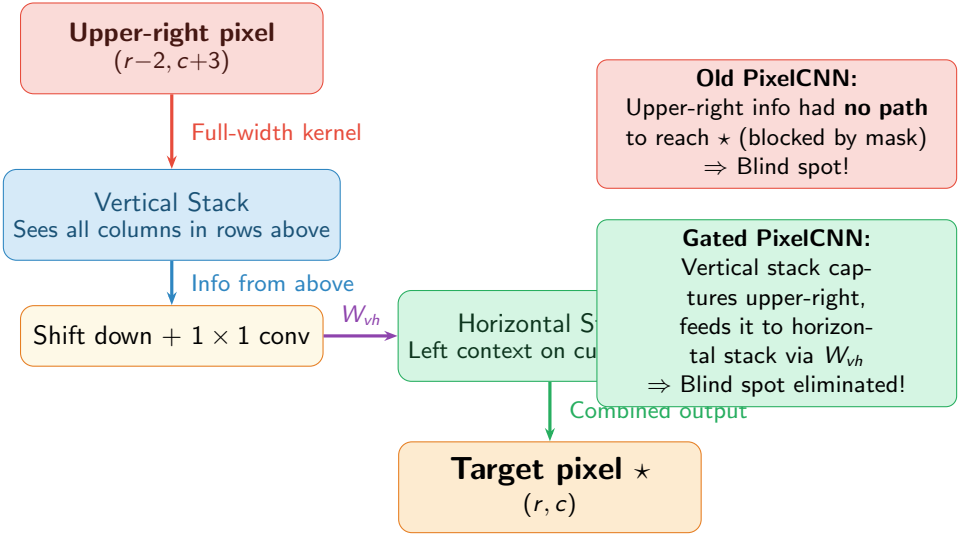
Per-layer computation:



$$\boxed{h = \mathrm{conv}(m_h \cdot w_h,\ x) + W_{vh} \cdot v_{\text{shifted}}}$$

The 1×1 conv merges vertical (above) and horizontal (left) information.

How Upper-Right Info Reaches ⋆



Why Autoregressive Property is Preserved

Three guarantees ensure pixel (r, c) never sees itself or future pixels:

1. Vertical stack

Shifted down 1 row
$\Rightarrow$ position (r, c)
only has info from
rows $< r$

2. Horizontal stack

$1 \times k$ mask
only looks left
$\Rightarrow$ position (r, c)
only has info from
columns $< c$
on row r

3. No upward path

Lower rows in both
masks are all zero
$\Rightarrow$ no information
flows from below

Combined: $\star$ at (r, c) only depends on pixels at (r', c') where $r' < r$, or $r' = r$ and $c' < c$.

This is exactly the **raster scan autoregressive ordering**.

Summary: PixelCNN $\rightarrow$ Gated PixelCNN

	PixelCNN	**Gated PixelCNN**
Structure	Single masked conv per layer	Vertical + Horizontal stacks
Mask shape	$k \times k$, upper-left only	Vertical: $k \times k$ upper rows Horizontal: $1 \times k$ left only
Receptive field	Sawtooth-shaped, has blind spot	Full coverage of all previous pixels
Key trick	—	Shift down + 1×1 conv connects stacks
Autoregressive	Yes (by mask)	Yes (by mask + shift)

Key Takeaway

Thank You

Questions?